

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



⚠ WARNING ⚠

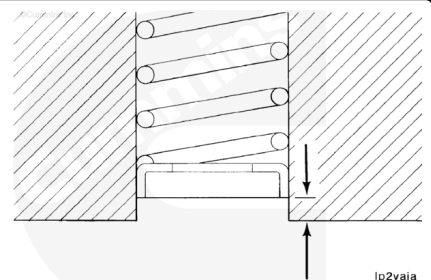
To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/35/35-007-037-tr.html>)
- Remove the oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/101/101-007-025-tr.html>)

Initial Check

Measure the installed depth of the relief valve retainer cup plug.

Relief Valve Retainer Installed Depth		
mm		in
8.03	MIN	0.316
8.53	MAX	0.336



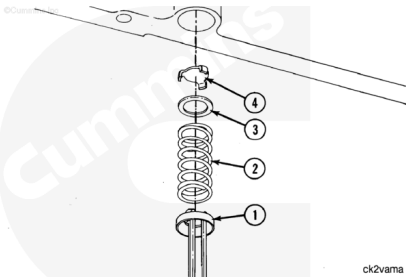
Note : If the plug depth is not within the specifications given, remove the high-pressure regulator assembly

and inspect the parts.

Remove

⚠ WARNING ⚠

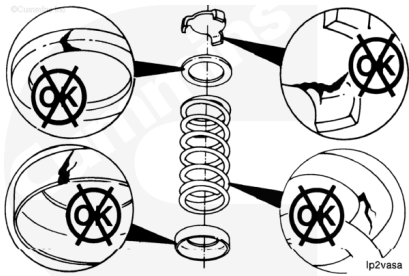
Use caution while removing the retainer plug (1). The pressure relief valve spring (2) is under compression. Personal injury can result. Wear face and eye protection.



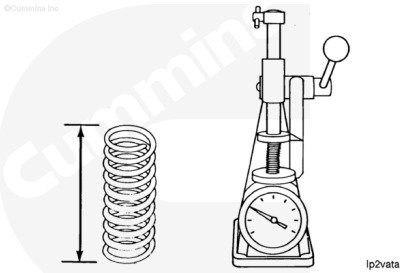
Use light-duty puller kit, Part Number 3375784 or equivalent, to remove the retainer plug (1).
Remove the pressure relief valve spring (2), washer (3), and valve disc (4).

Inspect for Reuse

Inspect the parts for damage.



Use a spring compression tester, Part Number 3375182, to measure the relief valve spring tension.
Compress the spring to 29.1-mm [1.15-in] height. The force required to compress the spring **must** meet these specifications.

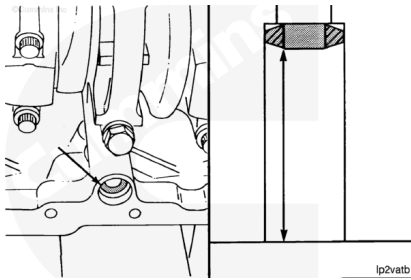


Force Required to Compress Spring to 29.1 mm [1.15 in]

n		lbf
263	MIN	59
322	MAX	72

Measure the relief valve seat depth in the cylinder block.

Relief Valve Seat Depth		
mm		in
62.62	MIN	2.465
63.62	MAX	2.505



If the relief valve seat is **not** at the specified depth or is loose or **not** level in the cylinder block, the seat **must** be removed and inspected. Use puller kit, Part Number 3375784, to remove the seat.

If the seating surfaces are free of indentations, the seat can be reused. If the surfaces have indentations, the seat **must** be replaced. Use relief valve seat driver, Part Number 3376486, to install the seat.

Install

⚠ WARNING ⚠

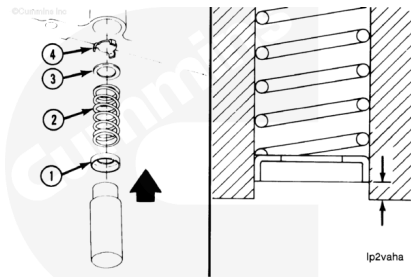
Use caution while installing the retainer plug (1). The pressure relief valve spring (2) is under compression. Personal injury can result. Wear face and eye protection.

Install the valve disc (4), washer (3) and spring (2).

Use flat washers equal to 5 mm [0.197 in] with plug driver, Part Number 3376489, to install the new retainer cup plug (1).

Measure the retainer plug depth in the cylinder block.

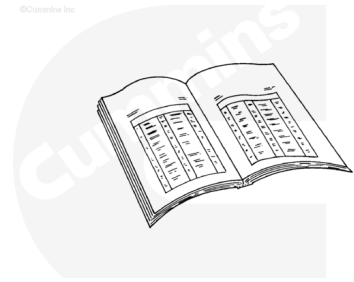
Relief Valve Retainer Installed Depth		
mm		in
8.03	MIN	0.316
8.53	MAX	0.336



Finishing Steps

⚠ CAUTION ⚠

Engine oil pressure must be indicated on the gauge within 15 seconds after starting. If oil pressure is not registered within 15 seconds, shut off the engine immediately to avoid engine damage. Confirm the correct oil level in the oil pan.



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- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/101/101-007-025-tr.html>)
- Fill the engine with clean 15W-40 lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/35/35-007-037-tr.html>)
- Operate the engine at idle speed for 3 minutes to inspect for leaks.